Navigation

Echolocation over long distances

Bats are known to use echolocation to orientate in their immediate environment, although whether they use a similar strategy for long-distance navigation (such as when flying towards their foraging grounds) remains unknown. Here, Goldshtein et al. show that bats can identify their location after being translocated several kilometres from their roost and are able to navigate home using echolocation alone, without relying on other senses.

The task of tracking echolocating bats — many of which weigh less than 10 grams — over long distances is technologically challenging. The authors used ATLAS, a lightweight, high-resolution GPS tracking system that enabled the tracking of Kuhl's pipistrelle bats (*Pipistrellus kuhlii*) after they were translocated ~3 km away from their roost. To investigate the sensory cues required for their subsequent navigation home, bats were split into groups with varying degrees of sensory deprivation, including control (normal senses); visually impaired; visual and magnetic deprivation; and visual, magnetic and olfactory deprivation. All of the bats were able to echolocate, and their navigation performance was evaluated on the basis of several metrics, including flight duration, trajectory and speed.

It was found that 95% of the bats successfully returned within a single night, independent of their sensory status, which indicates that bats can navigate large distances with echolocation alone. Notably, there was no difference in navigation performance between the various sensory deprivation treatments, although sighted (control) bats could navigate better than visually impaired bats, which indicates that navigation is improved when using both echolocation and vision. This finding is perhaps surprising, as Kuhl's pipistrelle bats have very small eyes with poor visual acuity, yet nonetheless they use visual information to assist navigation. Overall, these data provide direct evidence that bats can identify their location after translocation and fly home across long distances using echolocation.

Analysis of the flight angle relative to their destination revealed that the bats performed a meandering flight at the beginning of their journey and gradually transitioned to a direct flight towards their target. This steady improvement in navigation is consistent with the idea that bats use acoustic information to produce a cognitive map of their surroundings. To test this hypothesis, the authors generated a 3D landscape reconstruction of the study site and simulated echoes that each bat would receive along its journey. Their results revealed that when bats made movement decisions, they flew closer to environmental features with higher 'echoic entropy' — landmarks that provide more distinguishable acoustic information that might enable bats to recognize the objects and identify their location, an essential feature for map-based navigation.

Overall, these findings suggest that bats can navigate across kilometre-scale distances using echolocation to generate a cognitive map of their environment.

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Original article: Goldshtein, A. et al. Acoustic cognitive map–based navigation in echolocating bats. *Science* **386**, 561–567 (2024)

Motor systems

An unexpected role for macrophages in motor control

Mammals rely on sophisticated movements for surviving, and the stretch reflex serves as a fundamental mechanism for providing sensory feedback to generate adaptive responses. Muscle stretching activates specialized receptors located in skeletal muscles — the muscle spindles — that stimulate proprioceptive sensory afferents that directly synapse onto motor neurons, which drives muscle contraction. Since Sherrington's experiments in the late 19th century, the stretch reflex has been considered a closed-loop neuromuscular circuit. Here, Yan et al. find that, unexpectedly, muscle spindles contain an extra cellular component in macrophages that release glutamate and regulate muscle function.

The authors used a transcriptomic approach to determine differential gene expression between muscle spindle and non-spindle muscle tissue in mice. They found that muscle spindles showed an enrichment of neuronal genes (as expected) and, surprisingly, of immune cell markers. Further investigation by immunostaining revealed the presence of a population of macrophages inside the spindle capsule that not only had a unique molecular signature but also expressed genes more typically associated with neurons, such as those involved in glutamatergic excitatory neurotransmission.

The sensory afferent fibres of muscle spindles express glutamate receptors. To investigate whether muscle spindle macrophages (MSMPs) can regulate neural activity and muscle responses by releasing glutamate, the authors looked at the effects of optogenetically stimulating MSMPs in mouse hindlimb muscle. They found that this approach produced a rapid activation of primary sensory afferents and muscle contraction. Moreover, these effects were rapidly blocked by NMDA and AMPA-kainate receptor antagonists or by preventing SNARE-mediated vesicle release, which suggests that MSMPs regulate the stretch reflex circuit via glutamate release.

The findings described above have implications for locomotor performance, given the sensory-feedback role of this proprioceptive circuit. Indeed, the authors found (using transcriptomic analysis of dorsal root ganglion or muscle spindle primary afferents) that selective depletion of MSMPs from muscle tissue resulted in downregulation of several genes involved in locomotion. Compared with controls, the locomotor performance of mice with depleted or inactivated MSMP — including swimming and overground locomotion — was impaired. These mice showed lower levels of coordination and accuracy of motor responses, and a reduced ability to correct movement following a challenge.

Overall, these findings demonstrate that a population of macrophages that release glutamate — a function more commonly associated with neurons — directly regulate neural activity and muscle contraction by coupling muscle metabolic demands, sensation and action.

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Original article: Yan, Y. et al. Macrophages excite muscle spindles with glutamate to bolster locomotion. *Nature* https://doi.org/10.1038/s41586-024-08272-5 (2024)